

Python Programming Language

Python is a high-level, general-purpose programming language created by Guido van Rossum and first released in 1991. It emphasizes code readability through significant indentation and a clean, simple syntax.

Python supports multiple programming paradigms including procedural, object-oriented, and functional programming. It has a large standard library often described as having a batteries-included philosophy.

Python is widely used in web development, data science, artificial intelligence, scientific computing, and automation. Popular frameworks include Django and Flask for web development, and libraries like NumPy, Pandas, and TensorFlow for data science and machine learning.

The language uses dynamic typing and automatic memory management. Python's package manager, pip, allows developers to easily install and manage third-party libraries from the Python Package Index.

Python 3 was released in 2008 as a major revision that was not fully backward compatible with Python 2. Python 2 reached end of life in 2020, meaning Python 3 is now the standard version used in modern development.